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Urothelial Tumors of the Upper Urinary Tract and Ureter

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Keywords

transitional cell carcinoma; upper urinary tract; ureteral cancer renal pelvis cancer; urothelial carcinoma; UTUC

Questions

1. The highest incidence of upper urinary tract carcinomas (UTUC) occurs:
    a. in Western countries
    b. in Balkan countries
    c. in Western countries and Taiwan
    d. in Balkan countries and Taiwan
    e. Similarly in all countries
2. Smoking cessation results in:
    a. similar risk for 20 years
    b. reduction of 30% to 50% with interruption for more than 10 years
    c. reduction of 60% to 70% with interruption for more than 10 years
    d. reduction of 30% to 50% with interruption for more than 20 years

e. reduction of 60% to 70% with interruption for more than 20 years

3. UTUC related to blackfoot disease in endemic areas:
    a. is twice as common in males
    b. is twice as common in females
    c. is twice as common in males and related to younger onset of disease
    d. is twice as common in females and related to younger onset of disease
    e. both c and d
4. Inverted papillomas:
    a. are benign and only surveillance is necessary
    b. are benign but may harbor malignancy in 8% of the cases
    c. are benign but may harbor malignancy in 18% of the cases
    d. are divided into two types based on their morphology
    e. are divided into two types based on their histology
5. Micropapillary variant of urothelial carcinoma
    a. has a clinical outcome like a pure urothelial tumor
    b. is an independent predictor of progression-free survival
    c. is an independent predictor of advanced disease at diagnosis
    d. is an independent predictor of cancer-specific survival
    e. both c and d
6. Squamous cell carcinoma:
    a. is related to abuse of analgesics and chronic inflammation
    b. has an incidence 6 times higher in the renal pelvis than the ureter
    c. has an incidence 6 times higher in the ureter than the renal pelvis
    d. has an incidence 10 times higher in the renal pelvis than the ureter
    e. both a and b
7. Ureteroscopy:
    a. should be performed on all patients with urothelial cancer of the bladder
    b. does not increase the diagnostic accuracy with excretory or retrograde urography

c. increases the diagnostic accuracy with excretory or retrograde urography
d. both a and b
e. both a and c

8. The performance of biopsy during ureteroscopy:
   a. has an 82% correlation with low-grade tumors
   b. has a 92% correlation with high-grade tumors
   c. successfully detects 67% of invasive tumors
   d. a and b
   e. b and c
9. In ureteroscopic biopsy:
   a. lamina propria cannot be identified
   b. the tumor grade could be correlated to the radical surgery specimen in 50% of the cases
   c. muscle layer is present in the majority of the biopsies
   d. cup biopsies have the highest yield of lamina propria layer
   e. loop biopsies have the highest yield of lamina propria layer.
10. In CT urography, a filling defect related to a radiolucent stone could be distinguished by:
   a. measuring Hounsfield units (HU) in the range of 10 to 70 HU with an average 46 HU
   b. measuring Hounsfield units (HU) in the range of 70 to 100 HU with an average 82 HU
   c. measuring Hounsfield units (HU) in the range of 80 to 250 HU with an average 100 HU
   d. measuring Hounsfield units (HU) in the range of 250 to 850 HU with an average 420 HU
   e. none of the above
11. In individual studies, recurrence in the upper tract after treatment of urothelial bladder cancer:
   a. has an incidence of 2% to 6% and a median time to recurrence >3 years
   b. has an incidence of 2% to 6% and a median time to recurrence >5 years
   c. has an incidence of up to 10% and a median time to recurrence >3 years

d. has an incidence of up to 10% and an interval to recurrence of 17 to 170 months
e. has an incidence of up to 10% in most of the studies, but some report higher incidence up to 25%

12. Carcinoma in situ (CIS):
    a. increases the incidence of bilateral and multifocal disease
    b. is related to a likelihood of 25% for bilateral disease
    c. has an increased probability for panurothelial disease
    d. both a and c
    e. a, b, and c
13. Important preoperative risk factors include:
    a. older age
    b. tumor location
    c. continuation of tobacco consumption
    d. both a and c
    e. a, b, and c
14. Established postoperative risk factors include:
    a. lymphovascular invasion
    b. surgical margins
    c. tumor necrosis
    d. a and b
    e. a and c

Answers

1. **d. In Balkan countries and Taiwan.** The highest incidence of UTUC is observed in the Balkan countries and is related to the "Balkan nephropathy." The southeast coast of Taiwan also has a high prevalence of UTUC, which is related to the "Chinese herb nephropathy." Both nephropathies are associated with the presence of aristolochic acid in plants endemic in these countries.
2. **c. Reduction of 60% to 70% with interruption for more than 10 years.** McLaughlin et al. (1992) showed a 60% to 70% reduction in the risk for UTUC with interruption of smoking for more than 10 years.

3. **d. Is twice as common in females and related to younger onset of disease.** UTUC is twice as common in the female population in comparison to the worldwide incidence and is related to younger onset of disease in the areas endemic to blackfoot disease, such as the southwest coast of Taiwan. Blackfoot disease is related to chronic exposure to arsenic.
4. **c. Are benign but may harbor malignancy in 18% of the cases.** Inverted papillomas are considered benign. Nevertheless, studies have shown that 18% of these tumors of the ureter are malignant.
5. **b. Is an independent predictor of progression-free survival.** The micropapillary variant of the urothelial carcinoma was related to an independent predictor of progression-free survival (HR 3.85, $P = 0.003$) and to poorer cancer-specific survival than the nonmicropapillary UTUC ($P < .001$),
6. **e. Both a and b.** Squamous cell carcinoma has an incidence 5 times higher in the renal pelvis than in the ureter and has been associated with the abuse of analgesics and chronic inflammation.
7. **c. Increases the diagnostic accuracy with excretory or retrograde urography. Ureteroscopy increases the diagnostic accuracy of excretory or retrograde urography alone from 75% to 85%–90%.**
8. **e. b and c.** Biopsy during ureteroscopy has been reported to have a correlation with the pathohistological result of 92% for high-grade tumors and successfully detects 67% of the invasive tumors.
9. **e. Loop biopsies have the highest yield of lamina propria layer.** Loop biopsies may include the lamina propria layer in 100% of the specimens. The respective figure for cup biopsy is 62%.
10. **a. Measuring Hounsfield units (HU) in the range of 10 to 70 HU with an average 46 HU.** Urothelial cancers have an average density of 46 HU and range between 10 and 70 HU.
11. **a. Has an incidence of 2% to 6% and a median time to recurrence >3 years.** Upper tract recurrence after bladder cancer treatment has an incidence of 2% to 6% and is diagnosed later with a median time to recurrence >3 years. These rates have significant fluctuation when traditional series are considered. In most of the traditional series, the recurrence rate ranges between 2% and 4%

with an interval to recurrence ranging from 17 to 170 months, but there are also series with rates up to 25%.

12. **e. a, b, and c.** The presence of CIS increases the incidence of bilateral and multifocal disease. Bilateral disease has an overall incidence of 3% to 5% of UTUC cases. The presence of CIS increases this incidence to 25%. Patients with CIS have higher risk for subsequent panurothelial disease since there is a high probability for multifocal disease.
13. **e. a, b, and c.** Age is one of the most important demographic predictors of survival in UTUC. Decreased cancer-specific survival has been reported in patients of older age at the time of radical nephroureterectomy. Older age represents an independent factor for decreased survival. Being a smoker at diagnosis increases the risk for disease recurrence and mortality after radical nephroureterectomy and recurrence of the urothelial cancer in the bladder. Patients with ureteral and/or multifocal tumors seem to have a worse prognosis than those with renal pelvic tumors when the stage is adjusted.
14. **d. a and b.** Positive regional lymph nodes (LNs) is an independent factor for poor survival prognosis. Tumor stage remains the single most important factor for the oncological outcome of the UTUC patients.

Chapter Review

1. Upper urinary tract carcinomas comprise only 5% of the urothelial cancers.
2. The highest incidence is observed in individuals age 70 to 90 years in the Balkan countries, where the UTUC represents the 40% of all renal neoplasms.
3. Multifocal presence of UTUC is diagnosed in 10% to 20% of the cases.
4. Concurrent bladder cancer is diagnosed in 17% of the cases.
5. UTUCs are twice as frequent in men as in women.

6. Familial/hereditary UTUCs are linked to hereditary nonpolyposis colorectal carcinoma syndrome (Lynch syndrome).
7. Tobacco increases the relative risk for UTUC from 2.5 to 7.
8. UTUC “amino tumors” were related to occupational exposure to carcinogenic aromatic amines.
9. Balkan nephropathy is observed as a familial not inherited condition related to the dietary exposure to aristocholic acid.
10. Patients with pelvic and ureteral tumors have been reported to have a history of analgesic abuse in 22% and 11% of the cases, respectively.
11. The excess of inorganic arsenic in drinking water from artesian wells creates a significant risk for UTUC.
12. Urothelial carcinomas represent >90% of the upper urinary tract tumors. Pure nonurothelial upper urinary tract cancers are rare conditions.
13. Variants of urothelial cancer are encountered in approximately 25% of all urothelial cancers.
14. Papillomas, inverted papillomas, and Brunn’s nests are usually benign lesions.
15. UTUC develops through a gradual progression of hyperplasia to dysplasia and eventually CIS in a significant proportion of the UTUC cases.
16. CIS is difficult to diagnose with significant morphological variations.
17. The muscle invasion or invasion to the renal parenchyma or the surrounding adventitia is more likely to take place on the upper tract.
18. Localized disease: hematuria, dysuria, and flank pain.
19. Advanced disease: flank or abdominal mass, weight loss, anorexia, bone pain.
20. Cystoscopy should always be performed.
21. Flexible ureteroscopy with biopsy is a key approach for diagnosis.
22. Computed tomography urography has the highest diagnostic accuracy.
23. Urine collection or washing of the ureter with a ureteral catheter could provide the most accurate cytologic results.
24. Ureteral tumors are in the distal ureter in 70% of cases, midureter in 25% of cases, and upper ureter in 5% of cases. Bilateral disease occurs in 3% to 5% of cases.

25. Prognosis for recurrence after radical nephroureterectomy is related to the location of the tumor and stage.
26. The recurrence rate is 2% to 4%, with an interval of 17 to 170 months.
27. Bladder recurrence occurs in 15% to 75 % of cases within 5 years.
28. CIS: higher incidence of bilateral and multifocal disease.
29. Disease spread: direct expansion of the tumor, lymphatics, and blood stream.
30. The TNM classification and staging system. The 2004/2016 WHO grading classification.
31. Age alone should not prevent a curable approach.
32. Ureteral and/or multifocal tumors have a worse prognosis than renal pelvic tumors.
33. Muscle-invasive tumors have poor prognosis.
34. Positive regional LNs have poor survival prognosis.
35. Lymphovascular invasion is a predictor of disease recurrence and survival.

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Surgical Management of Upper Urinary Tract Urothelial Tumors

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Keywords

chemotherapy; immunotherapykidney-sparing surgery; nephroureterectomy; segmental ureterectomy; upper tract urothelial carcinoma

Questions

1. The majority of ureteral tumors occur in the:
    a. proximal ureter.
    b. midureter.
    c. distal ureter.
    d. proximal and midureter.
    e. distal and mid ureter.
2. The most important determinant of oncologic outcome in upper tract urothelial carcinoma is:
    a. stage and grade.
    b. number of tumors.
    c. location.
    d. tumor size.
    e. tumor architecture.
3. At the time of nephroureterectomy, the ureteral stump can be safely left in place for patients with urothelial tumors of the renal pelvis.

a. True
b. False

4. Initial evaluation of positive cytology should include which of the following?
    a. Cystoscopy
    b. Ureteroscopy
    c. CT urography
    d. a, b, and c
    e. a and b
    f. a and c

Pathology

1. A 38-year-old woman has right flank pain and microscopic hematuria. Cytology is atypical. CT scan shows a mass in the distal right ureter with hydronephrosis. Cystoscopy is negative, and attempted ureteroscopy is unsuccessful. The distal ureter is excised, and the pathology is depicted in Fig. 99.1 and is reported as endometriosis. The patient should be advised to:
    a. have a diagnostic laparoscopy.
    b. receive ablative hormonal therapy.
    c. have periodic upper tract imaging.
    d. have cystoscopy and cytology twice yearly for the next 2 years.
    e. have a hysterectomy and bilateral salpingo-oophorectomy.
2. A 60-year-old man has a right ureteral mass excised. The pathology is low-grade noninvasive transitional cell carcinoma (TCC) (Fig. 99.2). He has no prior history of upper tract disease or bladder cancer. Management should consist of:
    a. interval cystoscopies and cytology.
    b. instillation of BCG into the right upper tract.
    c. systemic platinum chemotherapy.
    d. periodic ureteroscopies of the left system.
    e. no further follow-up.

Answers

1. **e. Distal and midureter.** Ureteral tumors occur more commonly in the lower than in the upper ureter. **Overall, approximately 70% of ureteral tumors occur in the distal ureter, 25% in the midureter, and 5% in the proximal ureter.**
2. **a. Stage and grade.** The most well-established current predictors of survival in patients with upper tract urothelial tumors are stage and grade. The most significant decrease in survival is seen in T3 tumors, and higher-grade tumors are more likely to invade the surrounding tissues, hence presenting with higher stage. Although there have been studies showing differences in prognosis based on the tumor number, location, size, and architecture, these criteria are evolving and warrant further investigation.
3. **b. False. Complete removal of the distal ureter and bladder cuff offers oncologic outcomes superior to those for incomplete resection.** The risk of tumor recurrence in a remaining ureteral stump is 30% to 75%. In addition, adequate cystoscopic surveillance of a residual distal ureter stump after nephroureterectomy is virtually impossible, contributing to high rates of local recurrence. Therefore the entire distal ureter, including the intramural portion and the ureteral orifice, must be removed.
4. **f. a and c.** Ureteroscopy should be reserved for situations in which the diagnosis remains in question after conventional radiographic studies, and for patients in whom the treatment plan may be modified on the basis of the ureteroscopic findings. Although the risks of tumor seeding, extravasation, and dissemination are low in experienced hands, these risks are real and should preclude ureteroscopy when it is unnecessary. Because upper urinary tract tumors are often associated with bladder cancers, cystoscopy is mandatory in the evaluation to exclude coexistent bladder lesions.

Pathology

1. **c. Have periodic upper tract imaging.** With no symptoms, no further workup is indicated. The patient should be followed for possible development of a ureteral vesicle stricture at the site of the anastomosis.
2. **a. Interval cystoscopies and cytology.** Patients who have upper tract TCC have as high as a 30% incidence of bladder tumors and should be followed as for bladder cancer. Moreover, periodic upper tract imaging is necessary.

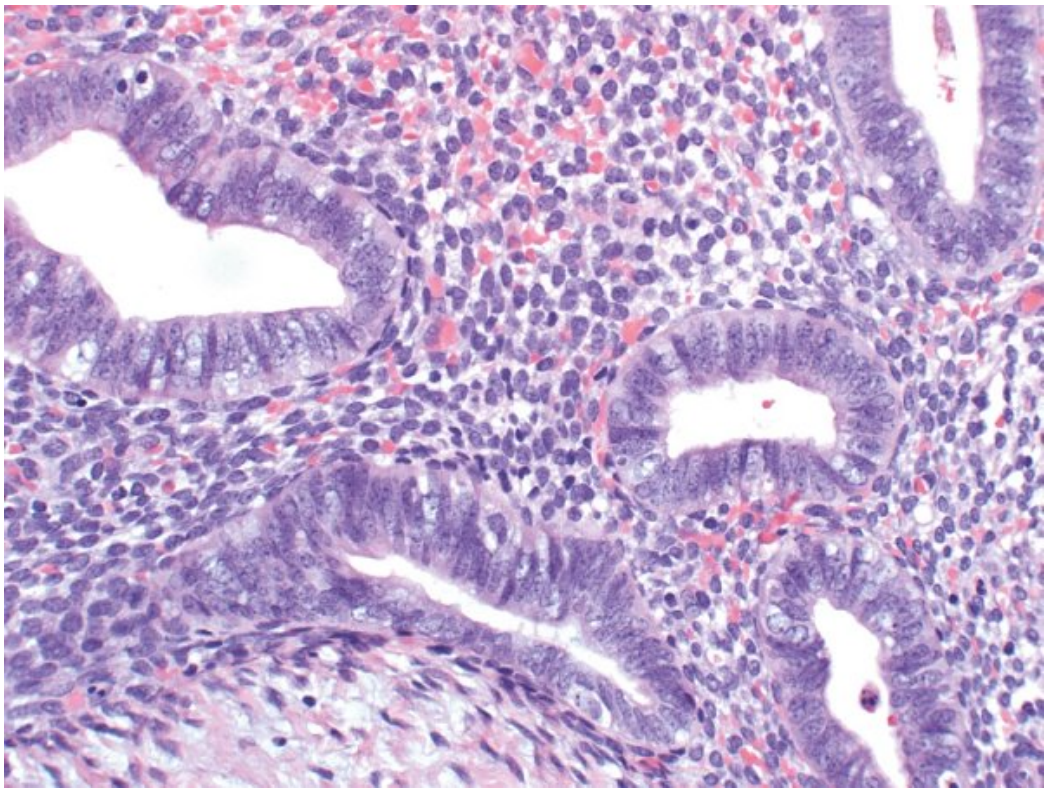


FIG. 99.1 (From Bostwick DG, Cheng L: *Urologic surgical pathology,* ed 2, Edinburgh, 2008, Mosby.)

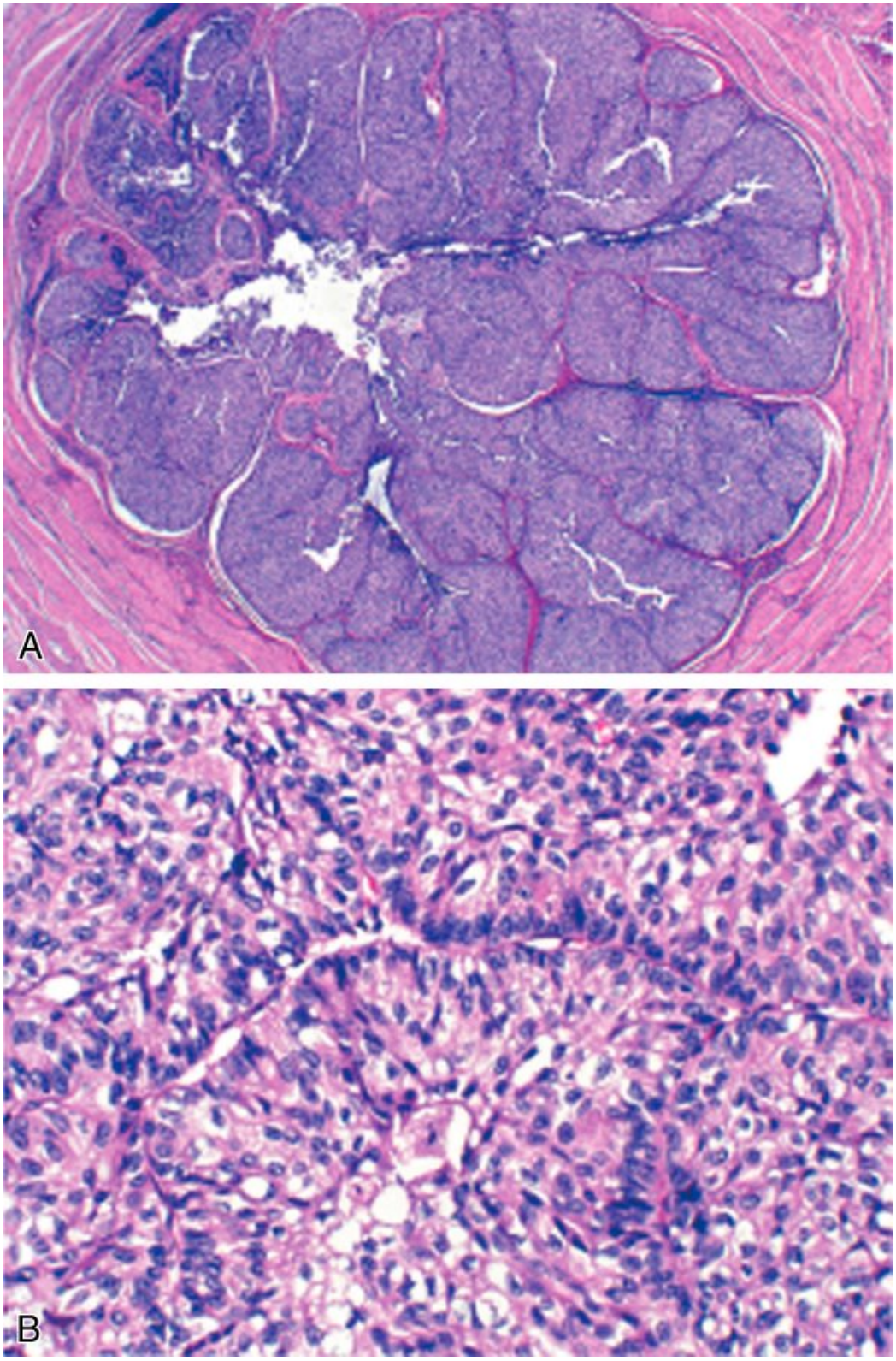


FIG. 99.2 (From Bostwick DG, Cheng L: *Urologic surgical pathology,* ed 2, Edinburgh, 2008, Mosby.)

Chapter Review

1. Bilateral upper tract tumors occur either synchronously or metachronously in 2% to 6% of patients.
2. In renal pelvic tumors, parenchyma invasion is the most significant predictor of metastases.
3. Inverted papillomas may be associated with upper tract tumors; it is not likely that cancers arise from them.
4. Squamous cell carcinoma and adenocarcinoma, although rare in the upper tract, are usually associated with long-term obstruction, inflammation, and occasionally calculi.
5. Tumor stage and grade, lymphovascular invasion, and lymph node spread are predictors of poor prognosis. The single most important predictor of outcome is stage.
6. A significant problem with ureteroscopic biopsy is that grade may be accurate, but accurate staging can be extremely difficult.
7. There is a 30% to 50% recurrence rate in ureteral tissue left distal to an invasive ureteral cancer.
8. Patients with T3 tumors located in the renal pelvis have a better survival than those with T3 tumors located in the ureter. Of renal pelvic tumors, 50% are invasive at diagnosis.
9. An adrenalectomy is not indicated for patients undergoing a nephroureterectomy for upper tract tumors.
10. After percutaneous resection of a tumor of the renal pelvis, the nephrostomy is left indwelling to allow for revisualization several weeks later to be certain that all tumors have been removed.
11. Lymphadenectomy has prognostic and possible therapeutic value in patients with T2 to T4 disease.
12. About 70% of ureteral tumors occur in the distal ureter, 25% in the midureter, and 5% in the proximal ureter.

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Retroperitoneal Tumors

Timothy A. Masterson, Clint Cary, and Richard S. Foster

Keywords

germ cell tumor; retroperitoneum; RPLND; sarcoma

Questions

1. Injury to the genitofemoral nerve will result in which neurologic manifestation?
    a. Loss of antegrade ejaculation
    b. Weakness with hip flexion of the lower extremity
    c. Paresthesias of the anterior thigh and lateral scrotal skin
    d. Erectile dysfunction
    e. None of the above
2. Liposarcomas of the retroperitoneum demonstrate all of the following characteristics EXCEPT:
    a. nonlipogenic components that are associated with more aggressive behavior
    b. overexpression of genes localized to chromosome 12q
    c. high frequency of local and regional recurrences
    d. excellent responsiveness to anthracycline-based chemotherapies for eradicating distant spread
    e. the need for both clinical and immunohistological characterizations to render an accurate diagnosis

3. Which of the following statements is true regarding solitary fibrous tumors?
    a. Can be associated with a paraneoplastic syndrome called Doege-Potter syndrome, which is characterized by hypoglycemia due to production of insulin-like growth factor
    b. Were previously referred to as hemangiopericytomas
    c. Are associated with uniform positivity for the *NAB2-STAT6* gene fusion product
    d. Demonstrate radiosensitivity
    e. All of the above
4. Among the spectrum of tumors classified as ganglioneuroblastomas, which of the following is true?
    a. Occur most commonly in elderly adults
    b. *N-myc* gene amplification is a negative prognostic marker
    c. Rarely, if ever, metastasize
    d. Survival rates exceed 90% at 5 years
    e. Unlike pheochromocytomas, metaiodobenzylguanidine (MIBG) scans have little value in staging disease
5. In the setting of a retroperitoneal cancer of unknown primary origin, all of the following are correct EXCEPT:
    a. PET-CT (positron emission tomography-computed tomography) offers little value in the initial workup of these patients
    b. Isochromosome i(12)p may identify patients with retroperitoneal primary germ cell tumors when assessing young, male patients
    c. Adenocarcinomas are the most common histologic subtype identified
    d. Represents up to 5% of new cancer cases worldwide
    e. Histology-specific chemotherapy is the mainstay of treatment
6. The incidence of perioperative acute respiratory distress syndrome (ARDS) in patients with prior receipt of bleomycin can be minimized by:
    a. avoidance of the Trendelenburg position
    b. keeping the FiO_2 as low as possible

c. short operating time
d. minimization of intra- and perioperative fluid resuscitation
e. both b and d

7. Where is the cisterna chylae located?
    a. Immediately posterolateral to the inferior vena cava (IVC) just superior to the right renal artery
    b. Immediately posterolateral to the aorta just superior to the left renal artery
    c. Immediately posterolateral to the aorta just inferior to the left renal artery
    d. Immediately posteromedial to the aorta just superior to the right renal artery.
    e. In the retrocrural region
8. When performing a full bilateral template retroperitoneal lymph node dissection (RPLND) for a mixed germ cell tumor in the retroperitoneum, the main packets are:
    a. paracaval, suprahilar, interaortocaval, ipsilateral gonadal vein
    b. retrocrural, interaortocaval, para-aortic
    c. paracaval, interaortocaval, para-aortic, interiliac
    d. paracaval, interaortocaval, para-aortic, ipsilateral gonadal vessels
    e. retrocrural, paracaval, para-aortic, interiliac
9. How many lumbar arteries need to be identified and divided to gain total vascular control of the abdominal aorta between the renal vessels and aortic bifurcation?
    a. 2
    b. 3
    c. 6
    d. 8
    e. 9

Answers

1. **c. Paresthesias of the anterior thigh and lateral scrotal skin.** The genitofemoral nerve is a sensorimotor nerve and originates from the

upper L1/2 segments of the lumbar plexus. It courses from the lumbar foramina and pierces the psoas major muscles anteriorly, traveling in a caudal direction toward the internal inguinal ring. It functions to provide sensation to the anterior thigh, lateral and anterior aspects of the scrotum in males, and mons pubis in female patients. The genital branch also has motor function, providing the innervation to the cremaster muscles in males.

2. **d. Excellent responsiveness to anthracycline-based chemotherapies for eradicating distant spread.** Liposarcomas represent soft tissue malignancies originating from adipocytes. Many tumors demonstrate aberrations of genes localized to chromosome 12q. The diagnosis of a liposarcoma can be difficult, and often requires clinical, radiographic, and histopathologic characterization to render it accurately. Surgical resection is the mainstay of treatment; however, local recurrences occur frequently in up to 40% of cases. Tumors with areas of de-differentiation are associated with more aggressive behavior, and metastatic disease responds poorly to systemic chemotherapy.
3. **e. All of the above.** All of the statements are true regarding solitary fibrous tumors. Previously referred to as hemangiopericytomas, they have been known to be associated with paraneoplastic syndromes due to endogenous insulinlike growth factor production. They are characteristically associated with the *NAB2-STAT6* gene product, which is diagnostic when present. Lastly, radiotherapy has been shown to improve surgical resection rates by decreasing positive margin rates.
4. **b. *N-myc* gene amplification is a negative prognostic marker.** Ganglioneuroblastomas are lethal tumors. While occurring most commonly in the first decade of life, they are associated with high rates of metastases. *N-myc* gene amplification has been recognized as a negative prognostic marker for disease progression. The 5-year survival rates are poor. MIBG scans and MRI have proven useful for staging of disease at time of diagnosis.
5. **a. PET-CT (positron emission tomography-computed tomography) offers little value in the initial workup of these patients.** Tumors of unknown primary origin represent a difficult

challenge. Initial diagnostics of value include the use of PET-CT and histologic characterization via biopsy to provide as much information to classify these entities. In young patients, the presence of i(12)p is pathognomonic for germ cell tumors, which are treated with platinum-based chemotherapy. Overall, this entity represents 5% of all new cancer cases in the world, with adenocarcinomas comprising 50% of these tumors.

6. **e. Both b and d.** Bleomycin-induced pulmonary toxicity is a well-described side effect of the drug, seen most commonly with a greater number of cumulative doses. Minimizing risk factors associated with acute respiratory toxicity and failure remains critical to the management of postchemotherapy patients in the perioperative period. Specifically, minimizing the amount of FiO_2 received as well as reducing perioperative fluid resuscitation have both been associated with a lower risk of acute exacerbation and decompensation.
7. **d. Immediately posteromedial to the aorta just superior to the right renal artery.** The cisterna chylae is most commonly identified in the posterior and medial position relative to the aorta cephalad in location to the right renal artery.
8. **d. Paracaval, interaortocaval, para-aortic, ipsilateral gonadal vessels.** After a series of modifications made to the original template for RPLND, exclusion of the suprahilar and interiliac regions were incorporated to reduce surgical morbidity without any significant risk for relapse in select patients. As such, the current standard for a bilateral template would include the para-aortic, paracaval, and interaortocaval lymph node packets. Additionally, the ipsilateral gonadal vessels should be removed concomitantly.
9. **c. 6.** Within the retroperitoneum, the infrahilar/infrarenal aorta has paired branches that extend posterolateral off of the aorta at regular intervals. Three such paired branches exist with few exceptions, for a total number of six lumbar artery branches that require prospective identification and division to achieve total vascular control of the aorta.

Chapter Review

1. Many tumors of the retroperitoneum (RP) grow silently over months or years, thus presenting with sizeable dimensions and adjacent organ involvement.
2. Cross-sectional imaging offers the best means of assessing size and extent of disease. Serologic testing plays an important role, specifically in germ cell tumors.
3. Fine-needle aspiration (FNA) and/or core biopsies remain an important tool in the evaluation and management of many diseases, and their diagnostic value should be considered in the initial workup of all RP masses.
4. Adjuvant and neoadjuvant therapies will continue to be expanded and incorporated as part of the multimodal approach to cancer treatment.
5. No disease process has exemplified this more than the management approach for testicular cancer patients.
6. In sarcoma, the heterogeneity across the spectrum of histologic entities and their biologic differences in response to radiation and systemic therapies restrict one's ability to make sweeping recommendations as to the timing and sequencing of different therapies.
7. Accordingly, using a multidisciplinary team to explore all available options to ensure optimal management of the RP mass offers patients the greatest chance of a successful outcome.

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Open Surgery of the Kidney

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Keywords

kidney; left renal vein; nephrectomy; right renal artery

Questions

1. A healthy 45-year-old man with no family history of cancer is found to have a 6-cm enhancing mass in the upper pole of his right kidney. A 2-cm solitary nodule is noted on preoperative chest radiography. Computed tomography (CT) confirms a solitary nodule in the lower lobe of the right lung. What is the most appropriate treatment course?
    a. Systemic chemotherapy alone
    b. Radical right nephrectomy and postoperative chemotherapy
    c. Biopsy of pulmonary nodule
    d. Radical nephrectomy and simultaneous pulmonary metastasectomy
    e. Radical nephrectomy with staged resection of pulmonary nodule 6 weeks postoperatively
2. What is the preferred technique for radical nephrectomy and removal of tumor thrombus above the level of the diaphragm in the absence of significant metastatic disease?
    a. Flank incision with extensive liver mobilization and removal of tumor through an incision in the diaphragm

b. Flank incision with cardiopulmonary bypass and deep hypothermic circulatory arrest (CPB-DHCA)
c. Chevron incision with CPB-DHCA
d. Chevron incision with Pringle maneuver
e. Midline incision with CPB-DHCA

3. Deep hypothermic circulatory arrest (DHCA) can have irreversible neurologic effects after what period of time?
   a. 10 minutes
   b. 20 minutes
   c. 40 minutes
   d. 60 minutes
   e. 90 minutes
4. In a 45-year-old man with a normal contralateral kidney and no family history of kidney cancer, in which of the following clinical scenarios would partial nephrectomy be indicated?
   a. Two tumors less than 3 cm each in the upper and lower pole
   b. Single 8-cm tumor in the upper pole
   c. Single 2-cm tumor in a hilar location with small renal vein tumor thrombus
   d. Single 4-cm tumor in any location
   e. All of the above
5. What is the strongest modifiable risk factor for renal insufficiency after partial nephrectomy?
   a. Duration of renal ischemia
   b. Surgical approach
   c. Administration of nephrotoxins
   d. Resection margin
   e. Administration of heparin
6. During a posterior right lumbotomy approach, what is the order of appearance of the renal artery, renal vein, and renal pelvis?
   a. Artery, renal pelvis, vein
   b. Artery, vein, renal pelvis
   c. Renal pelvis, artery, vein
   d. Vein, renal pelvis, artery
   e. Renal pelvis, vein, artery
7. Match the following T stage with the tumor characteristics.

1. T3c
2. T1a
3. T3a
4. T4
5. T2b

a. Greater than 10 cm confined to capsule
b. Less than 4 cm confined to capsule
c. 6 cm invading adrenal gland
d. 5 cm with renal sinus fat invasion
e. 13 cm with renal vein thrombus invading the wall of the inferior vena cava

8. Five days after left partial nephrectomy for a hilar tumor, there is persistent drainage from the Penrose drain site. Laboratory analysis of the drain fluid demonstrates elevated amylase levels. Imaging studies demonstrate small bowel dilation consistent with ileus and fluid around the tail of pancreas. What is the ideal management?
   a. Antibiotics
   b. Immediate surgical exploration
   c. Percutaneous drain placement
   d. Nasogastric tube placement, parenteral nutrition, and conservative management
   e. Nasogastric tube placement, low-fat diet, and conservative management
9. Which segmental branch of the renal artery is most consistent and supplies 25% of the arterial supply to the renal unit?
   a. Apical (superior) segmental artery
   b. Anterior superior segmental artery
   c. Posterior segmental artery
   d. Anterior inferior segmental artery
   e. The basilar (inferior) segmental artery
10. What maneuver refers to the reflection of the second and third portions of the duodenum in a medial direction to expose the right renal vessels and ventral inferior vena cava?
    a. Cattell maneuver
    b. Langenbeck maneuver
    c. Sorcini maneuver

d. Kocher maneuver
e. Pringle maneuver

11. What partial nephrectomy technique should be used as a last resort in a solitary kidney?
    a. Enucleation
    b. Wedge resection
    c. Cryotherapy
    d. Polar resection
    e. Extracorporeal repair and autotransplantation
12. The subcostal nerve may be inadvertently transected during an anterior subcostal incision for a radical nephrectomy. Between what two layers does this nerve run?
    a. Posterior peritoneum and transversalis fascia
    b. Scarpa fascia and external oblique muscle
    c. External oblique and internal oblique
    d. Internal oblique and transversalis
    e. Skin and Scarpa fascia
13. What is the motor deficit resulting from transaction of the subcostal nerve?
    a. Winged scapula
    b. Hemidiaphragmatic paralysis
    c. Paresis of the flank musculature and flank bulge
    d. Inability to flex ipsilateral adductor muscle
    e. Weakness of contralateral rectus abdominis muscle
14. What percentage of patients have multiple renal arteries?
    a. 0% to 2%
    b. 2% to 10%
    c. 10% to 20%
    d. 20% to 30%
    e. More than 30%
15. Which of the following is NOT an indication for simple nephrectomy?
    a. Nonfunctional chronically infected kidney
    b. Nonfunctional persistently hydronephrotic kidney causing pain

c. Renovascular hypertension refractory to medical and nephron-sparing surgical intervention
d. Polycystic kidney with minimal function and recurrent infections
e. Kidney with 8-cm enhancing upper pole hilar mass

16. Two days after cardiopulmonary bypass and circulatory arrest (20 minutes) for an extensive right-sided renal mass with thrombus extending into the atrium, using traditional median sternotomy, a relatively healthy 36-year-old patient is unable to be extubated and has no purposeful right-sided movement. Imaging reveals a large left-sided cerebrovascular infarct. What clinical scenario can explain this event?
    a. Pulmonary air embolism
    b. Cerebral ischemia from bypass and circulatory arrest
    c. Tension pneumothorax
    d. Right main stem bronchial intubation
    e. Unrecognized paradoxical embolism
17. Which form of therapy has been considered the gold standard for localized renal cell carcinoma?
    a. Chemotherapy
    b. Immunotherapy
    c. Radiation
    d. Hormonal therapy
    e. Surgical resection
18. On postoperative day 2 after radical nephrectomy for a 14-cm complex left renal tumor using an anterior midline incision, there are overt signs of peritonitis. The patient is 72 years old with significant atherosclerotic disease. At exploration, the entire small bowel is necrotic and nonviable. What artery was inadvertently ligated?
    a. Celiac
    b. Left gastric
    c. Inferior mesenteric
    d. Superior mesenteric
    e. Right gastroepiploic

19. During resection of a large right renal mass, the main renal artery is identified, ligated, and divided, but the renal vein fails to decompress. What is the most likely explanation for this?
    a. Renal vein tumor thrombus
    b. Subclinical renal arteriovenous malformation
    c. Bleeding disorder
    d. Arterial collateral branch vessels
    e. Extensive venous collateral obstruction
20. What is most appropriate setting for a thoracoabdominal incision?
    a. Large right upper pole renal mass with tumor thrombus in the renal vein
    b. 5-cm right renal tumor in a hilar location
    c. Large left lower pole tumor with extensive lymphadenopathy
    d. Large right renal mass with tumor thrombus to the retrohepatic level
    e. A 10-cm right lower pole tumor with arteriovenous malformation
21. What is the most common complication associated with performing CPB-DHCA for the removal of large renal cell tumor thrombus?
    a. Pulmonary air emboli
    b. Intestinal ischemia
    c. Bleeding and coagulopathy
    d. Lower extremity tumor emboli
    e. Tumor emboli
22. Which of the following is NOT a proposed benefit of renal artery embolization (RAE)?
    a. Shrinkage of an arterialized tumor thrombus to ease surgical removal
    b. Reduced blood loss
    c. Facilitation of dissection due to tissue plane edema
    d. Ability to ligate the renal vein before the renal artery at time of nephrectomy
    e. Modulation of the immune response
    f. None of the above
23. What is the most common complication after RAE?
    a. Groin hematoma from puncture site

b. Paraplegia from spinal artery occlusion
c. Coil migration
d. Postinfarction syndrome (pain, nausea, and fever)
e. Adrenal insufficiency

24. What is the most common complication after partial nephrectomy for nonexophytic renal masses?
    a. Hemorrhage
    b. Renal failure
    c. Rhabdomyolysis
    d. Hydronephrosis
    e. Urinary leak
25. Ten days after a left partial nephrectomy for a 4.5-cm hilar tumor, there is persistent fluid output from the surgical drain. No ureteral stent was placed at the time of surgery, and a small opening in the collecting system was oversewn. The creatinine concentration of the drain fluid is 34.5 mg/dL, consistent with urine. Despite conservative management, the volume fails to decline. A retrograde pyelogram demonstrates a moderate amount of contrast extravasation, confirming the urinary fistula. What is the most appropriate management at this time?
    a. Immediate reexploration and repair
    b. Percutaneous nephrostomy tube placement
    c. Removal of surgical drain
    d. Internalized ureteral stent placement
    e. Internalized ureteral stent placement, continued surgical drain monitoring, and placement of Foley catheter

Answers

1. **d. Radical nephrectomy and simultaneous pulmonary metastasectomy.** This patient would be best managed with a radical nephrectomy and simultaneous removal of the pulmonary nodule. Systemic therapy is not a primary treatment unless there is extensive metastatic disease at presentation. Given his age and lack of medical problems, there is no reason to delay the removal of his

kidney and the pulmonary nodule. The tumor location and pulmonary nodule both can be accessed through one incision (i.e., thoracoabdominal).

2. **c. Chevron incision with CPB-DHCA.** CPB-DHCA has been established as the most prudent course for the removal of these tumor thrombi. The chevron incision provides the best exposure. Alternatives to CPB, including extensive liver mobilization and intrapericardial resection, carry an increased risk of bleeding.
3. **c. 40 minutes.** The duration of DHCA can vary depending on the degree of tumor thrombus. Vena cava resection and substitution can add additional time if there is significant tumor invasion into the wall of the vena cava. Studies have suggested that irreversible neurologic effects may be observed after 40 minutes of DHCA.
4. **d. Single 4-cm tumor in any location.** In patients with a normal contralateral kidney, the current literature supports elective partial nephrectomy for single T1 tumors.
5. **a. Duration of renal ischemia.** Duration of renal ischemia is the strongest modifiable risk factor for renal insufficiency after partial nephrectomy.
6. **c. Renal pelvis, artery, vein.** The renal pelvis is the first structure one encounters with the posterior right lumbotomy incision, followed by the artery and vein. This approach can be used to repair ureteropelvic junction obstruction, especially in children or patients with multiple prior abdominal and/or flank surgeries.
7. a: T2b; b: T1a; c: T4; d: T3a; e: T3c.
8. **d. Nasogastric tube placement, parenteral nutrition, and conservative management.** Conservative management of a pancreatic fistula should be the first approach in this patient. Initial nasogastric tube placement can help resolve the ileus. Parenteral nutrition will limit any pancreatic secretions from oral intake.
9. **c. Posterior segmental artery.** The posterior division is the first and most consistent branch point of the renal artery and supplies approximately one-fourth of the blood supply.
10. **d. Kocher maneuver.** Mobilization of the second and third portions of the duodenum is referred to as a *Kocher maneuver.* The Pringle maneuver is the temporary occlusion of the porta hepatis.

The Langenbeck maneuver is the division of the coronary and right triangular ligaments, providing medial rotation of the right lobe of the liver and exposure of the suprarenal inferior vena cava.

11. **e. Extracorporeal repair and autotransplantation.** All patients with solitary kidneys are high-risk candidates for partial nephrectomy and may have transient renal impairment postoperatively. The degree and duration of renal impairment may be increased owing to risks associated with renal autotransplantation (hemorrhage, thrombosis, lymphocele, stenosis).
12. **d. Internal oblique and transversalis.** The subcostal nerve runs between these two layers. Caution must be taken not to sever this nerve during flank incisions.
13. **c. Paresis of the flank musculature and flank bulge.** Damage to the subcostal nerve results in denervation and paresis of the flank musculature, leading to chronic postoperative pain or flank bulge.
14. **d. 20% to 30%. Multiple postmortem and radiographic studies estimate that 25% of the general population have supernumerary renal arteries.**
15. **e. Kidney with 8-cm enhancing upper pole hilar mass.** There should be little reservation about performing a radical nephrectomy for an enhancing mass, especially in the upper pole. Almost all nonmalignant disease affecting the kidney can be treated via a simple approach.
16. **e. Unrecognized paradoxical embolism.** This rare but devastating clinical situation occurs in patients with a patent foramen ovale. An embolism may originate from tumor thrombus manipulation or from deep venous thromboembolism.
17. **e. Surgical resection.** There have been numerous studies to suggest that surgical resection is the mainstay of therapy for kidney cancer.
18. **d. Superior mesenteric.** Ligation of the superior mesenteric artery produces ischemia in the bowel distribution above. **The superior mesenteric artery can be mistaken for the left renal artery from the anterior approach. Visualizing the artery from a posterior position as it enters the hilum will help to minimize this complication.**

19. **d. Arterial collateral branch vessels.** Failure of the renal vein to decompress after ligation of the main renal artery indicates additional arterial inflow, which may be secondary to a missed lower or upper pole artery or extensive collateral arteries.
20. **a. Large right upper pole renal mass with tumor thrombus in the renal vein.** The thoracoabdominal incision is ideal for larger tumors involving the upper pole. The incision is also ideal for managing tumor thrombus extending into the renal vein. The inferior vena cava can be nicely exposed via this approach.
21. **c. Bleeding and coagulopathy.** Intraoperatively, the administration of heparin in addition to hypothermia leads to significant coagulopathy. The bleeding from heparin is typically limited to an "ooze" intraoperatively and should not consume time and energy during the operation. After tumor removal, the rewarming process helps to promote coagulation.
22. **f. None of the above. Proposed benefits of preoperative RAE include shrinkage of an arterialized tumor thrombus to ease surgical removal, reduced blood loss, facilitation of dissection due to tissue plane edema, ability to ligate the renal vein before the renal artery at time of nephrectomy, and modulation of the immune response.**
23. **d. Postinfarction syndrome (pain, nausea, and fever). The triad of fever, flank pain, and nausea occurs in up to 75% of patients after angioembolization.** Fevers can often exceed 39.4°C (103°F) and are best managed with antipyretics.
24. **e. Urinary leak.** Partial nephrectomy for nonexophytic masses has an increased risk of entering the collecting system. Even when the collecting system is closed under direct vision, there may still be extravasation of urine that collects in the perirenal space. The use of postoperative surgical drains is imperative in the management of these collections to reduce the risk of infections. In addition, the drain output volume can be observed to determine if collections are resolving. Renal failure is rare unless operating on a solitary kidney or on a patient with marginal renal function. Rhabdomyolysis can be encountered secondary to patient positioning and increased body mass index.

25. **e. Internalized ureteral stent placement, continued surgical drain monitoring, and placement of Foley catheter. Placement of a ureteral stent can promote urine drainage into the bladder. Keeping a Foley catheter in place reduces urine reflux.**

Chapter Review

1. The right renal artery is posterior to the inferior vena cava.
2. Renal arteries are end arteries; ligation results in infarction of the segment that they supply.
3. The renal venous network intercommunicates.
4. Lumbar veins often enter the left renal vein and, not infrequently, the right renal vein. They enter posteriorly. Care must be taken when encircling the renal vein not to tear one of these lumbar veins.
5. There is no conclusive evidence that renal artery embolization has any immunologic therapeutic benefit.
6. The renal artery is always ligated before the renal vein when performing a nephrectomy; each vessel is ligated individually.
7. Patients with a glomerular filtration rate of less than 60 mL/min or those with significant proteinuria are at risk for postoperative renal failure following renal surgery—particularly when a nephrectomy is performed.
8. Adrenalectomy is not recommended as part of a radical nephrectomy unless imaging shows adrenal involvement with tumor or an upper pole tumor is contiguous with the adrenal.
9. Transesophageal echocardiography is an excellent modality to determine the level of the vena cava tumor thrombus immediately before the surgical event.
10. In patients with vena cava tumor thrombi cephalad to the hepatic venous outflow who require CPB, either mild hypothermia and no circulatory arrest or significant hypothermia with circulatory arrest may be performed. Each technique has its advantages and disadvantages. The method used is at the discretion of the surgeon.
11. The addition of a lymphadenectomy to a radical nephrectomy for renal cell carcinoma has a questionable impact on progression-free

and overall survival. It may be considered in patients who have enlarged lymph nodes on preoperative imaging, those in whom cytoreductive surgery is being performed, and those with ominous pathologic findings of the primary renal tumor.

12. Ligation of the right renal vein will result in failure of the right renal unit due to lack of venous collateral vessels.
13. Ligation of the left renal vein is possible because collateral venous drainage may occur through lumbar and gonadal vessels.
14. The renal vein ostium of the vena cava should be excised in patients with vena cava tumor thrombi, because invasion of the vena cava vein wall at this site is not uncommonly found.
15. Twenty-five percent of the general population have supernumerary renal arteries.
16. The superior mesenteric artery can be mistaken for the left renal artery from the anterior approach. Rarely, the hepatic artery can be mistaken for the right renal artery. Visualizing the artery from a posterior position relative to the renal vein as it enters the hilum will help identify the renal artery.
17. Proposed benefits of preoperative renal artery embolization include shrinkage of an arterialized vena cava tumor thrombus to ease surgical removal, reduced blood loss, facilitation of dissection due to tissue plane edema, and the ability to ligate the renal vein before the renal artery at time of nephrectomy. These patients may develop the postinfarction syndrome (pain, nausea, and fever). The triad of fever, flank pain, and nausea occurs in up to 75% of patients after angioembolization.

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Laparoscopic and Robotic Surgery of the Kidney

Daniel M. Moreira, and Louis R. Kavoussi

Keywords

calyceal diverticulectomy; complications; kidney; laparoendoscopic single-site surgery; laparoscopy; natural orifice transluminal endoscopic surgery; nephrectomy; nephrolysis; nephropexy; partial nephrectomy; radical nephrectomy; renal biopsy; robot-assisted; surgery

Questions

1. The following statements are true regarding laparoscopic and robotic renal surgery in obese patients, EXCEPT:
    a. Obesity is associated with distorted anatomy and limited range of instrument motion.
    b. Morbid obesity (body mass index [BMI] >40) is a contraindication to laparoscopic or robotic surgery.
    c. Complication rates are higher among obese patients.
    d. Conversion to open surgery is more common in obese patients.
    e. Longer surgical instruments are sometimes required to complete the procedure.
2. The following statements are true regarding laparoscopic and robotic surgical approaches to renal surgery, EXCEPT:
    a. Transperitoneal is the most widely used approach.

b. Retroperitoneal partial nephrectomy is best suited for anterior renal tumors.
c. Disadvantages of retroperitoneal approach include the limited working space and decreased instrument triangulation.
d. The advantages of robotic approach include greater degree of instrument motion, elimination of tremor, and the ability to scale motions.
e. Advantages of transperitoneal approach include familiar anatomic landmarks and more flexibility of trocar placement.

3. The following statements are true regarding laparoendoscopic single site (LESS) and natural orifice transluminal endoscopic surgery (NOTES) compared to conventional laparoscopic renal surgery, EXCEPT:
    a. For both LESS and NOTES, conventional laparoscopic instrumentation cannot be used, and flexible or articulated instruments are required.
    b. LESS consolidates all laparoscopic ports in a single skin incision.
    c. LESS and NOTES are associated with improving cosmesis and reduced postoperative pain.
    d. LESS has limited triangulation due to the clustering of instruments entering the intracorporal working space.
    e. NOTES uses one or more patent natural orifices of the body for access.
4. The following are indications for renal cysts disease treatment, EXCEPT:
    a. cyst-associated pain.
    b. ureteral obstruction.
    c. suspicion for malignancy.
    d. infection.
    e. autosomal dominant polycystic kidney disease.
5. The following statements are true regarding laparoscopic and robotic renal surgery for renal cystic disease, EXCEPT:
    a. Symptomatic benign cysts can be initially managed with percutaneous needle aspiration.

b. Multiple or bilateral cysts should not be treated in the same operation.
c. Laparoscopic management of symptomatic renal cysts is associated with greater than 90% symptom improvement.
d. If a sclerosing agent is used, it should not be injected into the collecting system.
e. Most renal cysts do not require surgical intervention.

6. The following statements are true regarding laparoscopic renal biopsy, EXCEPT:
   a. It is indicated in cases of hematuria and proteinuria of glomerular origin and/or unexplained renal failure.
   b. The most common complication is bleeding.
   c. Ultrasound guidance can facilitate the localization of the kidney in cases of abundant retroperitoneal and perinephric adipose tissue.
   d. A 5-mm biopsy forceps or an 18-gauge Tru-Cut needle can be used for tissue sampling.
   e. It is contraindicated in cases of multiple renal cysts or solitary kidney.
7. The following statements are true regarding nephropexy and nephrolysis, EXCEPT:
   a. Nephrolysis consists of the renal mobilization with skeletonization of the hilar vessels and upper ureter.
   b. Nephrolysis is an effective treatment for severe refractory chyluria.
   c. Nephropexy is indicated in cases of nephroptosis associated with pain or vascular or urinary obstruction.
   d. Nephropexy can be accomplished with either suture fixation or the use of foreign material, such as mesh or tissue adhesive.
   e. Laparoscopic nephropexy is associated with greater than 80% resolution of nephroptosis but almost no pain improvement.
8. The following statements are true regarding calyceal diverticulectomy, EXCEPT:
   a. Most cases of calyceal diverticulectomy require hilar clamping.

b. Indications for surgical intervention include chronic pain, recurrent urinary tract infection, gross hematuria, and decline in renal function.
c. Calyceal diverticulectomy involves widening of the infundibular drainage to prevent urinary stasis or complete ablation of the diverticula cavity.
d. Injection of dye via a preoperatively placed externalized ureteral catheter is helpful in localizing the diverticulum.
e. Laparoscopic, robotic, or LESS transperitoneal or retroperitoneal approaches can be used.

9. The following are advantages of laparoscopic over open radical nephrectomy, EXCEPT:
a. improved cosmesis.
b. shorter hospital stay.
c. improved cancer-specific survival.
d. less analgesic use.
e. shorter convalescence time.

10. Which of the following is an absolute contraindication to laparoscopic radical nephrectomy?
a. Tumor size greater than 14 cm
b. Age older than 80 years
c. Uncorrected coagulopathy
d. Renal vein thrombus
e. Multiple prior abdominal surgeries

11. Regarding laparoscopic radical nephrectomy, the following statement is CORRECT:
a. En bloc hilar vessel stapling is unsafe and should not be attempted.
b. The use of a retrieval bag is not needed in cases of tumor extraction after morcellation.
c. Metastatic renal cell carcinoma is an absolute contraindication.
d. Clips alone are contraindicated for the ligation of the renal artery.
e. Adrenal gland should be bluntly hand-dissected away from the upper pole during hand-assisted nephrectomy.

12. Compared to traditional laparoscopic nephrectomy, hand-assisted nephrectomy is associated with higher:
    a. local recurrence.
    b. wound-related complications.
    c. pain scores.
    d. conversion to open surgery.
    e. costs.
13. Regarding minimally invasive partial nephrectomy, the following statement is CORRECT:
    a. Central and hilar tumors are a contraindication to laparoscopic partial nephrectomy.
    b. Compared to open partial nephrectomy, robot-assisted partial nephrectomy in solitary kidney is associated with higher conversion to radical nephrectomy.
    c. The placement of a drain is generally considered unnecessary when the incision in the pelvicalyceal system is repaired with running suture.
    d. Suture renorrhaphy is the most reliable technique to achieve hemostasis of the partial nephrectomy bed.
    e. Multiple ipsilateral renal tumors should not be removed in the same setting.
14. Regarding warm renal ischemia and hilar control, the following statement is INCORRECT:
    a. Hilar control can be accomplished by using bulldog clamps or a laparoscopic Satinsky clamp.
    b. Renal artery–only clamping is a safe alternative to artery and vein clamping.
    c. Parenchymal compression is best suited for peripheral tumors.
    d. Off-clamp partial nephrectomy is a suitable option for the resection of small, exophytic, and noninfiltrating tumors.
    e. Hilar unclamping before the completion of the renorrhaphy is contraindicated.
15. Regarding laparoscopic thermal ablation, the following statement is INCORRECT:
    a. Compared to partial nephrectomy, the advantages of thermal ablation include less blood loss, short operative time, and

fewer complications.

b. The advantages of the laparoscopic approach over percutaneous ablation include direct tumor visualization, mobilization of the kidney, and dissection and retraction of surrounding structures.

c. Contact of the iceball with the renal collecting system, ureter, renal vasculature, or adjacent organs should be avoided.

d. Intraoperative monitoring of radiofrequency ablation is based on direct temperature measurement or measurement of electrical impedance.

e. Temperatures greater than 105°C throughout the tumor are recommended for an adequate thermal damage.

16. Regarding complications of minimally invasive renal surgery, the following statement is INCORRECT:

a. Advanced age is a major risk factor for complications.

b. Partial nephrectomy is associated with higher complication rates compared to radical nephrectomy and simple nephrectomy.

c. Large tumors, obesity, and prior abdominal surgery are associated with a higher complication rate.

d. Chronic pyelonephritis, such as xanthogranulomatous pyelonephritis, is associated with greater rate of conversion to open surgery.

e. The most common postoperative complications are bleeding, wound infection, pneumonia, and prolonged ileus.

17. Regarding complications of minimally invasive renal surgery, the following statement is INCORRECT:

a. Most of the life-threatening intraoperative complications occur during the access phase and dissection of the renal hilum.

b. Most cases of urinary leak require ureteral stenting or percutaneous nephrostomy placement.

c. Mild to moderate venous bleeding can be controlled by applying direct pressure.

d. Factors associated with urinary leak include large tumor size, hilar location, pelvicalyceal repair, and prolonged warm ischemia time.

e. Stapler malfunction is associated with a 20% conversion rate to open surgery and 2% mortality rate.

18. Regarding complications of minimally invasive renal surgery, the following statement is INCORRECT:
    a. Persistent and increased trocar site pain without significant erythema or purulent drainage is a common sign of visceral injury.
    b. Splenic injuries are most commonly caused by vigorous traction on splenic ligaments.
    c. Gallbladder injuries are best managed by primary surgical repair.
    d. Capsular pancreatic injuries can be managed by closing the defect with nonabsorbable suture and/or drain placement.
    e. Most liver injuries can be managed with topical hemostatic therapy and argon beam coagulation.
19. A 55-year-old woman presents to the emergency department on postoperative day 5 after laparoscopic left radical nephrectomy complaining of vomiting, fever, and diffuse abdominal pain. Computed tomography (CT) scan with oral contrast shows leakage of contrast from the ascending colon. The most likely cause of this injury is:
    a. bowel ischemia.
    b. trocar placement.
    c. blunt dissection.
    d. electrocautery.
    e. abdominal wall closure.
20. A 71-year-old woman presents to emergency department with new-onset gross hematuria on postoperative day 6 after an uncomplicated laparoscopic right partial nephrectomy. Her blood pressure is 90/56 mm Hg and heart rate is 112 beats/min. Hematocrit is 21%. After fluid resuscitation, her vital signs normalized, but the hematuria worsens and requires continuous bladder irrigation. The best next step is:
    a. cystoscopy and retrograde pyelogram.
    b. radical nephrectomy.
    c. renal angiography and embolization.

d. admission to intensive care unit and blood transfusion.
e. CT scan with intravenous contrast.

Answers

1. **b. Obesity is associated with distorted anatomy and limited range of instrument motion.** Although morbid obesity is associated with distorted anatomy, limited range of instrument motion, and higher complication and conversion rates, it is not a contraindication to laparoscopic or robotic renal surgery. In general, obese patients benefits from minimally invasive surgery given it provides less incisional pain and shorter convalescence.
2. **b. Retroperitoneal partial nephrectomy is best suited for anterior renal tumors.** Although anterior renal tumors can be resected with retroperitoneal partial nephrectomy, this technique is best suited for posterior tumors. A study comparing 100 transperitoneal and 63 retroperitoneal laparoscopic partial nephrectomies found that 77% of the posterior tumors were managed via retroperitoneal approach and 97% of the anterior tumors were managed with transperitoneal surgery.
3. **a. For both LESS and NOTES, conventional laparoscopic instrumentation cannot be used, and flexible or articulated instruments are required.** Although LESS has limited triangulation due to the clustering of instruments entering the intracorporeal working space, conventional laparoscopic instrumentation can be used. Flexible or articulated instruments are helpful but not required in cases of LESS or NOTES.
4. **e. Autosomal dominant polycystic kidney disease.** Cyst-associated pain, urinary obstruction, suspicion for malignancy, and infection are traditional indications for operative management of renal cysts. Autosomal dominant polycystic kidney disease alone is not an indication for surgical intervention unless it is accompanied by complications, pain, suspicion for malignancy, etc.
5. **b. Multiple or bilateral cysts should not be treated in the same operation.** Although most renal cysts do not require surgical

intervention, multiple or bilateral cysts can and frequently are treated in the same operation, with greater than 90% symptom improvement and radiologic resolution.

6. **e. It is contraindicated in cases of multiple renal cysts or solitary kidney.** Although renal biopsy is typically done percutaneously under CT or ultrasound guidance, there are circumstances where a laparoscopic biopsy may be preferred, including failed previous attempts at percutaneous biopsy, renal anatomic abnormalities, bleeding diathesis, morbid obesity, multiple renal cysts, or a solitary kidney.
7. **e. Laparoscopic nephropexy is associated with greater than 80% resolution of nephroptosis but almost no pain improvement.** Laparoscopic nephropexy is associated with a 90% to 100% pain improvement, 80% to 95% resolution of the 5-cm renal descent, and 90% improvement in renal function measured by nuclear medicine scan. Nephrolysis consists of the renal mobilization with skeletonization of the hilar vessels and upper ureter and is an effective treatment for severe refractory chyluria.
8. **a. Most cases of calyceal diverticulectomy require hilar clamping.** Given most cases of calyceal diverticulectomy have low bleeding risk, hilar control is usually not required.
9. **c. Improved cancer-specific survival.** Advantages of laparoscopic radical nephrectomy include improved cosmesis, shorter hospital stay, less analgesic requirement, and shorter convalescence time. Oncologic outcomes including recurrence, progression to metastatic disease, and cancer-specific survival are comparable with open radical nephrectomy.
10. **c. Uncorrected coagulopathy.** Uncorrected coagulopathy is an absolute contraindication to laparoscopic renal surgery, as are untreated infection and hypovolemic shock. Laparoscopic partial nephrectomy has been reported as a feasible and safe approach in the management of select large renal masses, elderly patients, renal vein thrombosis, and multiple prior abdominal surgeries.
11. **d. Clips alone are contraindicated for the ligation of the renal artery.** Although previously there have been concerns of arteriovenous fistula formation in cases of en bloc hilar vessel

stapling, this approach has been shown to be a safe alternative. The use of a retrieval bag is mandatory in cases of tumor extraction after morcellation to prevent tumor spillage. Cytoreductive laparoscopic nephrectomy is safe and effective in patients with metastatic renal cell carcinoma. The hand should not be used to bluntly dissect the adrenal gland free from the upper pole because this will typically result in bleeding. Clips alone are contraindicated for the ligation of the renal artery due to reports of fatal cases of clip failure.

12. **b. Wound-related complications.** Hand-assisted laparoscopic nephrectomy offers recovery, morbidity and cost that are comparable with those of pure laparoscopy surgery. Recurrence-free, cancer-specific, and overall survivals are similar between surgical techniques. The rate of wound-related complications such as wound infection and incisional hernia are slightly higher in hand-assisted approach compared to pure laparoscopy.
13. **d. Suture renorrhaphy is the most reliable technique to achieve hemostasis of the partial nephrectomy bed.** Minimally invasive partial nephrectomy for central and hilar tumors can be performed safely by an experienced surgeon with perioperative outcomes comparable with those of peripheral tumors. Minimally invasive partial nephrectomy for tumors in a solitary kidney is safe and offers comparable outcomes to open surgery. Repeat renal surgery can pose a technical challenge and is associated with higher blood loss and complication rates compared to first-time renal surgery. Although the laparoscopic resection of multiple tumors is associated with long operative time and hospitalization, in experience hands it is associated with comparable blood loss, complication rates, and renal functional outcomes as resection of a solitary tumor. The placement of a drain is usually recommended in most cases of pelvicalyceal system, even cases that are repaired with running sutures, to reduce the risk of perinephric urinoma. Although there are other means of achieving hemostasis of the partial nephrectomy bed such as the use of clips and tissue sealants, suture renorrhaphy is considered the most reliable technique to achieve hemostasis.

14. **e. Hilar unclamping before the completion of the renorrhaphy is contraindicated.** During radiofrequency ablation, thermal damage occurs through frictional heating due to ionic oscillation by a monopolar or bipolar high-frequency alternating current which can induce temperatures between 60°C and 100°C throughout the tumor. Temperatures greater than 105°C should be avoided to prevent tissue vaporization.
15. **e. Temperatures greater than 105° C throughout the tumor are recommended for an adequate thermal damage.** Hilar control can be accomplished by using bulldog clamps on the renal artery with or without venous clamping, or a laparoscopic Satinsky clamp for en bloc hilar clamping. Studies evaluating early unclamping (before the completion of the renorrhaphy) found this technique to be associated with shorter warm ischemia time and comparable estimated blood loss and complication rates.
16. **a. Advanced age is a major risk factor for complications.** Advanced age alone is not a major risk factor for complications. Obesity, comorbidities, prior abdominal surgery, and larger and more anatomically complex tumors have all been associated with higher complication rates.
17. **b. Most cases of urinary leak require ureteral stenting or percutaneous nephrostomy placement.** Factor associated with urinary leak include large tumor size, hilar location, pelvicalyceal repair, and prolonged operative and warm ischemia time. Most cases of urinary leak require no intervention other than percutaneous drainage. When conservative management fails or there is evidence of distal obstruction, additional intervention may be required, such as ureteral stenting and bladder decompression.
18. **c. Gallbladder injuries are best managed by primary surgical repair.** Gallbladder injuries are best managed by cholecystectomy.
19. **d. Electrocautery.** Although bowel injury can occur at any point during the procedure, the most common case of unrecognized injury is the use of thermal energy adjacent to the bowel, responsible for nearly 50% of the cases. Thermal injuries usually present later than nonthermal bowel injuries.

20. **c. Renal angiography and embolization.** Significant gross hematuria after partial nephrectomy suggests postoperative bleeding from an arteriovenous fistula or pseudoaneurysm. Although a CT scan with intravenous contrast may help to diagnose and localize a postoperative bleed, this will ultimately delay treatment. Immediate renal angiogram with selective arterial embolization allows for diagnosis, localization, and definitive treatment in the same setting.

Chapter Review

1. Minimally invasive renal surgery provides less incisional pain, shorter convalescence, and better cosmesis compared to open surgery, with comparable efficacy and long-term outcomes.
2. The indications for minimally invasive renal surgery are generally similar to those for open renal surgery.
3. The basic principles of oncologic surgery, including complete tumor excision with negative margins associated with maximization of renal parenchyma preservation when feasible, should be followed when using minimally invasive approaches to treat renal tumors.
4. Familiarity with the anatomic landmarks, surgical techniques, and instrumentation and knowledge of each procedure and its potential pitfalls are key for successful minimally invasive surgery.

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Nonsurgical Focal Therapy for Renal Tumors

Chad R. Tracy, and Jeffrey A. Cadeddu

Keywords

cryoablation; image-guided surgery; kidney cancer; minimally invasive surgery; radiofrequency ablation

Questions

1. Nephron-sparing surgery (partial nephrectomy):
    a. is the predominant treatment modality used in the United States for the management of small renal masses.
    b. offers equivalent cancer-specific and overall survival compared with radical nephrectomy.
    c. when performed laparoscopically, it is associated with fewer complications than radiofrequency ablation (RFA) or cryoablation.
    d. offers metastatic recurrence-free survival and cancer-specific survival similar to that of radiofrequency ablation and cryoablation.
    e. was not impacted by the advent of hand-assisted laparoscopic radical nephrectomy.
2. With regard to deciding on a particular ablation modality, which of the following should be considered?

a. RFA is associated with less postoperative hemorrhage than cryoablation.
b. Both RFA and cryoablation are subject to “heat sink.”
c. The American Urological Association (AUA) guidelines currently recommend consideration of ablation for primary treatment of T1a renal tumors, even for younger, healthier patients, with the understanding that recurrence rates may be higher and salvage procedure more difficult.
d. All of the above.
e. Both a and b are true.

3. The critical treatment temperature threshold during cryoablation at which irreparable cell damage is achieved is:
a. 0°C.
b. −60°C.
c. −20°C.
d. −40°C.
e. −19.4°C.

4. Critical parameters for successful renal cryosurgery include:
a. a double freeze-thaw cycle.
b. achieving a critical target temperature.
c. treatment under real-time image guidance.
d. treatment to beyond 1 cm of the targeted lesion.
e. all of the above.

5. Compared with renal cryoablation, the primary disadvantage of RFA is:
a. higher risk of hemorrhage following RFA.
b. inability to use RFA laparoscopically.
c. inability to monitor treatment under image guidance.
d. inferior cancer-specific survival.
e. none of the above.

6. Recent meta-analyses have demonstrated:
a. higher local recurrence rates with ablative technologies when compared with partial or radical nephrectomy.
b. no evidence for the superiority of one ablation technology versus the other with regard to oncologic outcomes or complications.

c. a lack of uniformity regarding evaluation, patient selection, treatment, and follow-up of patients undergoing renal tumor ablation.
d. fewer complications with renal tumor ablation compared with extirpative treatments.
e. all of the above.

7. The most important technique to increase the size of RF lesions is:
a. using higher RF currents.
b. applying RF currents faster to achieve better heating.
c. clamping hilar vessels.
d. reducing impedance by improving current conductivity. using "dry" RFA.

8. Compared to radiofrequency ablation and cryoablation, newer technologies may have which of the following advantages?
a. High-intensity focused ultrasound (HIFU) requires no percutaneous needle placement, yet results in equivalent oncologic outcomes.
b. Irreversible electroporation (IRE) has a potential advantage for treating central tumors due to its nonthermal mechanism that preserves tissue scaffolding, large blood vessels, and urothelium.
c. Microwave ablation works through a similar mechanism to RFA and is therefore similarly limited by tissue charring and heat sink.
d. Stereotactic body radiation therapy (SBRT) is highly effective at treating renal cell carcinoma but is limited due to mobility of the kidney with respiration.
e. Routine use of selective embolization has consistently demonstrated improved outcomes for ablation of all renal mass, regardless of location.

9. Following tumor ablation, the most reliable method of documenting treatment success is:
a. biopsy of the treatment area with hematoxylin and eosin (H&E) staining.
b. biopsy of the treatment area with reduced nicotinamide adenine dinucleotide (NADH) diaphorase staining.

c. follow-up computed tomography (CT) or magnetic resonance imaging (MRI) with contrast that demonstrates complete loss of contrast enhancement and stable or decreased size of the treated area.
d. follow-up CT or MRI without contrast that demonstrates a decrease in size of the treated area.
e. all of the above.

Answers

1. **d. Offers metastatic recurrence-free survival and cancer-specific survival similar to that of radiofrequency ablation and cryoablation. Nephron-sparing surgery is considered the gold-standard treatment for small renal masses but is underused in the United States.** Partial nephrectomy offers equivalent oncologic outcomes when compared with radical nephrectomy and demonstrates superior renal functional preservation. Laparoscopic partial nephrectomy offers excellent cancer-specific end points but is technically challenging and is associated with significant complications. Cryoablation and radiofrequency ablation are associated with significantly fewer complications than laparoscopic partial nephrectomy. **There is no significant difference in metastatic recurrence-free survival between extirpative and ablative treatments. Cancer-specific survival (CSS) following RFA is comparable to laparoscopic partial nephrectomy. CSS is comparable between open partial nephrectomy and both RFA and cryoablation.** Preliminary studies suggest that the technical ease and minimal morbidity of hand-assisted laparoscopic nephrectomy may ultimately lead to fewer nephron-sparing procedures.
2. e. Both a and b are true.
3. **d. − 40°C.** Experimental evidence suggests that irreversible cellular damage is achieved in normal renal parenchyma at −19.4°C. However, tumor cells require lower treatment temperatures to

achieve uniform cellular necrosis. The recommended treatment temperature during renal cryosurgery is −40°C.

4. **e. All of the above.** Animal studies have demonstrated that a single freeze-thaw cycle is inferior to a double freeze-thaw cycle with respect to adequacy of tissue ablation and local tumor control. As mentioned, complete cellular necrosis is consistently achieved at a targeted temperature of −40°C. Campbell and colleagues (1998) [a] demonstrated that the aforementioned threshold temperature of −40°C was achieved 3.1 mm inside the edge of the evolving ice ball. **To guarantee that the tumor is completely ablated with a margin of normal tissue, the ice ball is generally carried 5 to 10 mm beyond the edge of the tumor when viewed under real-time imaging.**
5. **c. Inability to monitor treatment under image guidance.** The primary disadvantage when employing RFA for the treatment of renal lesions is difficulty in monitoring treatment under real-time image guidance. The risk of hemorrhage is higher with cryoablation. RFA may be employed laparoscopically, percutaneously, or openly. **Cancer-specific survival is equivalent between RFA and cryoablation.**
6. **e. All of the above.** Three recent meta-analyses evaluated outcomes following partial nephrectomy, radical nephrectomy, cryoablation, and RFA for the treatment of small renal masses. **Both RFA and cryoablation demonstrated higher local recurrence rates when compared with partial or radical nephrectomy.** However, the majority of published studies on cryoablation and RFA have enrolled small numbers of patients and used disparate operative and follow-up protocols. The pathology is either not obtained or is difficult to interpret, and the reliability of radiographic imaging remains unknown. **Although the aggregate duration of follow-up is too short to derive irrefutable conclusions, treatment outcomes with cryoablation and RFA appear comparable.** In general, complications occur with equal or less frequency with tumor ablation when compared with partial or radical nephrectomy.

7. **d. Reducing impedance by improving current conductivity.** The electrically conductive agent facilitates the delivery of energy from the electrode to surrounding tissue, rendering it a much larger, "virtual" electrode. Impedance remains lower, and larger volumes are ablated. One technique to accomplish this is to employ saline-cooled electrodes, which reduce charring at the tip of the electrode and thereby reduce impedance.
8. **b. Irreversible electroporation (IRE) has a potential advantage for treating central tumors due to its nonthermal mechanism that preserves tissue scaffolding, large blood vessels, and urothelium.** IRE is produced through a series of electrical pulses delivered by a single (bipolar) or multiple (monopolar) electrodes. With appropriate modulation it is able to ablate a substantial and reproducible amount of tissue by increasing cell membrane permeability that ultimately leads to cell death. Because it is nonthermal, studies have confirmed its ability to spare the urothelium and larger blood vessels.
9. **c. Follow-up computed tomography (CT) or magnetic resonance imaging (MRI) with contrast that demonstrates complete loss of contrast enhancement and stable or decreased size of the treated area.** Complete loss of contrast enhancement on follow-up CT or MRI has been considered a sign of complete tissue destruction. **Following cryoablation, the treated area demonstrates a decrease in size of approximately 50% in the year following treatment. Following RFA, the treated area may not demonstrate a decrease in size. Any increase in size following treatment should be viewed as an ominous sign, and intervention should be directed accordingly.** Postablative biopsy, although helpful, yields ambiguous results following RFA. Its use and interpretation remain controversial.

Chapter Review

1. Currently, a double freeze-thaw cycle is suggested for cryotherapy with a period of time during each cycle in which the temperature is

maintained at −40°C.

2. Radiofrequency waves generate heat by causing ionic agitation in the tissue through which they pass; charring at the probe tip prevents adequate conduction of the wave from the probe into tissue; to prevent heat at the probe becoming excessive with resultant charring, the probes are cooled.
3. CT or MRI is used to follow patients after ablative therapy. There should be a complete lack of contrast enhancement at the tumor site. Successful cryoablation on follow-up imaging often reveals up to a 50% reduction in size of the lesion. RF ablation followed over the long term often shows a slight reduction in size and cavitation.
4. The tumor should be biopsied to confirm its malignant nature before ablative therapy is performed.
5. The most common complication following cryoablation is hemorrhage; a serious complication following radiofrequency ablation is ureteral/collecting system injury. Bleeding is less common following radiofrequency ablation.
6. High-intensity focused ultrasonography generates heat by focusing the ultrasound waves at a point; unfortunately, skin burns are a common complication of this technology.
7. There is no significant difference in metastatic recurrence-free survival between extirpative and ablative treatments. Cancer-specific survival (CSS) following RFA is comparable to laparoscopic partial nephrectomy. CSS is comparable between open partial nephrectomy and both RFA and cryoablation.

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[a] Sources referenced can be found in *Campbell-Walsh Urology*, 12th Edition, on the Expert Consult website.

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Treatment of Advanced Renal Cell Carcinoma

Ramaprasad Srinivasan, and W. Marston Linehan

Keywords

axitinib; bevacizumab; cabozantinib; chromophobe; clear cell; collecting duct; everolimus; FLCN; fumarate hydratase; interleukin-2; levantinib; medullary renal cancer; MET; nivolumab; papillary; pazopanib; RCC; renal cell carcinoma; sorafenib; succinate dehydrogenase; sunitinib; temsirolimus; TFE3; TFEB; VHL

Questions

1. What is the approximate overall objective response rate to interleukin-2 (IL-2) monotherapy in patients with metastatic renal cell carcinoma (RCC)?
    a. 5%
    b. 10%
    c. 15%
    d. 25%
    e. 35%
2. Which of the following regarding IL-2 therapy for metastatic RCC is TRUE?
    a. IL-2 has demonstrable efficacy in clear cell as well as papillary RCC.

b. Randomized studies have demonstrated a survival benefit associated with high-dose IL-2.
c. Low-dose subcutaneous and high-dose intravenous IL-2 have comparable efficacy.
d. Durable complete responses are seen in a small proportion of patients receiving high-dose IL-2.
e. Newer formulations have led to better tolerability of high-dose IL-2.

3. In the Memorial Sloan-Kettering Cancer Center (MSKCC) prognostic scheme for patients with metastatic RCC undergoing therapy with cytokine or chemotherapy, which of the following is NOT a predictor of poor outcome?
   a. Karnofsky performance status greater than 80%
   b. Elevated lactate dehydrogenase
   c. Elevated calcium
   d. Decreased hemoglobin
   e. Interval from diagnosis to systemic therapy initiation of <1 year
4. A 50-year-old man with a 12-cm left renal renal mass, identified on a CT scan to evaluate abdominal pain, is referred for further management. He is otherwise healthy except for mild fatigue that has developed over the past 3 months; laboratory evaluation reveals a hemoglobin of 9.8 mg/dL and an elevated corrected calcium (10.8 mg/dL) but an otherwise normal chemistry panel and complete blood count. A CT scan of the chest reveals multiple pulmonary nodules consistent with metastatic disease. A biopsy of the renal mass reveals clear cell RCC, Fuhrman grade 3. Which of the following is true with regard to surgical management of this patient?
   a. Cytoreductive nephrectomy should be performed since this patient appears to be otherwise healthy with a good performance status.
   b. Systemic therapy should be offered to this patient since he is unlikely to benefit from cytoreductive nephrectomy.
   c. Sytemic therapy should be commenced, with cytoreductive nephrectomy after six cycles of sunitinib.

d. The patient has high-risk disease and should commence therapy with everolimus.

5. Which of the following statements about cytokine therapy for metastatic renal cell carcinoma is TRUE?
   a. Lymphokine-activated killer (LAK) cells augment the efficacy of both interferon-α (IFN-α) and IL-2.
   b. Randomized trials have demonstrated a significant survival advantage for combined IL-2 and interferon versus either agent given as monotherapy.
   c. The combination of IL-2 and interferon leads to higher overall response rates than either agent alone.
   d. The complete response rate with interferon-α monotherapy is 10%.
6. Which of the following metastatic RCC tumors is most likely to benefit from cytokine therapy?
   a. Papillary carcinoma
   b. Clear cell carcinoma
   c. Medullary carcinoma
   d. Collecting duct carcinoma
   e. Chromophobe carcinoma
7. A 58-year-old woman had a nephrectomy 6 years previously for a grade 2 clear cell carcinoma. She was incidentally found to have three left-sided pulmonary nodules (two <1.0 cm, other 2.5 cm). A physical examination is normal, as are all blood chemistries. Computed tomography (CT) of the brain, lungs, abdomen, and pelvis show three pulmonary nodules with no associated hilar or mediastinal adenopathy, and a bone scan is normal. Which of the following is the most appropriate next step in her management?
   a. Therapy with high-dose IL-2
   b. Biopsy of a pulmonary nodule
   c. Mediastinoscopy followed by resection of the pulmonary nodules
   d. Observation
   e. IFN-α therapy
8. The sirolimus analogs temsirolimus and everolimus act primarily on which of the following pathways?

a. Vascular endothelial growth factor (VEGF)
b. Platelet-derived growth factor (PDGF)
c. Raf-1
d. Mechanistic target of rapamycin (mTOR)
e. C-met

9. The overall response rate in metastatic clear cell RCC patients receiving front-line therapy nivolumab in combination with ipilimumab is:
   a. 15% to 20%.
   b. 30% to 40%.
   c. 60% to 70%.
   d. less than 10%.
   e. greater than 70%.
10. A newly diagnosed patient with metastatic clear cell RCC is referred for discussion of systemic therapy options. You determine that he has intermediate risk disease based on standard prognostic criteria and that systemic therapy should be commenced. Which of the following decisions would be supported by data from randomized phase 3 studies?
    a. Sunitinb is the most appropriate choice for initial treatment of this patient
    b. Initiate therapy with a combination of nivolumab and ipilimumab since it is associated with better response rates and overall survival compared with sunitinib
    c. Initiate therapy with a combination of pazopanib and nivolumab
    d. Initiate therapy with a combination of lenvatinib and everolimus
11. Which of the following agents has been shown to prolong progression-free survival in patients with metastatic clear cell RCC who have progressed on first-line therapy with VEGFR antagonists?
    a. Axitinib
    b. Bevacizumab + IFN-α
    c. High-dose IL-2
    d. Bevacizumab

e. Low-dose subcutaneous IL-2

12. Randomized trials in patients with previously untreated metastatic clear cell RCC have demonstrated that:
    a. the combination of pembrolizumab and axitinib is associated with better overall survival compared to sunitinib.
    b. the combination of avelumab in combination with axitinib is associated with better progression free survival compared to sunitinib.
    c. cabozantinib is associated with a better response rate and progression free survival compared to sunitinib.
    d. all of the above.
    e. none of the above.
13. Which of the following agents has been shown in randomized phase 3 trials to prolong survival in "poor-risk" metastatic RCC patients?
    a. IL-2
    b. Sunitinib
    c. Sorafenib
    d. Temsirolimus
    e. IFN-α
14. Which of the following molecules is not known to be upregulated as a consequence of VHL inactivation in clear cell RCC?
    a. VEGF
    b. PDGF
    c. TGF-α
    d. Glut-1
    e. Raf-1
15. In what proportion of sporadic clear cell tumors are mutations or promoter hypermethylation of the *VHL* gene seen?
    a. 70% to 90%
    b. 10% to 20%
    c. 100%
    d. Less than 10%
16. A 47-year-old man presents with multiple metastatic lesions to the lungs and liver 8 months following a radical nephrectomy for a 9-

cm papillary type I renal tumor. Which of the following statements about his systemic treatment options is TRUE?

a. Sunitinib is associated with a 30% to 40% overall RECIST response rate in this subtype of RCC.
b. Sorafenib is associated with better long-term outcomes than sunitinib in papillary type I RCC.
c. mTOR inhibitors improve survival in patients with metastatic papillary RCC.
d. Enrollment in a phase 2 trial evaluating a novel inhibitor of MET activity is a reasonable consideration in this patient.

Answers

1. Although response rates of 30% or more were reported in early phase 2 studies with IL-2, the overall response rate with this agent was determined to be approximately 15% in larger studies and meta-analysis.
2. **d. Durable complete responses are seen in a small proportion of patients receiving high-dose IL-2.** Complete responses are seen in 7% to 9% of metastatic clear cell RCC patients receiving high-dose IL-2, with the majority of these remaining disease-free for more than 15 years. **The efficacy of IL-2 has not been adequately evaluated in patients with non–clear cell histologies, and the use of this agent is largely restricted to clear cell RCC patients.** There are no randomized phase 3 studies demonstrating survival benefit with IL-2. While IL-2 use is limited in current clinical practice, it may be considered in well-selected patients because of its ability to provide durable responses in some patients.
3. **a. Karnofsky performance status greater than 80%.** A Karnofsky performance score below 80% was determined to be an adverse prognostic feature and is one of the factors used to predict outcome in the MSKCC prognostic system for patients with metastatic RCC. All other factors listed have been associated with poor outcome.

4. **b. Systemic therapy should be offered to this patient since he is unlikely to benefit from cytoreductive nephrectomy.** This patient has multiple poor prognostic indicators and is not likely to benefit from cytoreductive nephrectomy. While data from a randomized phase 3 study conducted in the era of cytokine therapy, as well as retrospective data in patients receiving therapy with VEGFR-targeted agents, suggest a benefit for cytoreductive nephrectomy in well-selected patients, a recent phase 3 study that randomized newly diagnosed intermediate and poor-risk patients with metastatic clear cell RCC to receive either cytoreductive nephrectomy followed by sunitinib or sunitinib alone (CARMENA) demonstrated no benefit for surgery preceding systemic therapy. There are insufficient data to support routine cytoreductive nephrectomy following suntinib; a study evaluating this approach (SURTIME) was closed early because of slow accrual. Everolimus is not appropriate in the front-line setting.
5. **c. The combination of IL-2 and interferon leads to higher overall response rates than either agent alone.** In a randomized phase 3 study, the combination of IL-2 and interferon was associated with a higher response rate than either agent given alone, although this did not translate to an improved long-term outcome (overall survival) in the combination arm. **The addition of LAK cells to cytokine therapy does not appear to improve outcome.**
6. **b. Clear cell carcinoma.** Clear cell RCC is the histology most likely to respond to cytokine therapy. Although there are inadequate data to make definitive determinations about the activity of cytokines such as IL-2 or interferons in other histologic subtypes, these agents do not appear particularly effective in non–clear cell RCC variants.
7. **c. Mediastinoscopy followed by resection of the pulmonary nodules.** Patients with metachronous pulmonary nodules related to remote renal tumors may have prolonged survival following resection of the nodule(s) as demonstrated by several retrospective studies.
8. **d. Mechanistic target of rapamycin (mTOR).** mTOR (mammalian target of rapamycin) is the primary target of sirolimus (rapamycin)

and its analogs.

9. **b. 30% to 40%.** Objective overall response rates in metastatic clear cell RCC patients treated with nivolumab in combination with ipilimumab was 42% in a randomized phase 3 study comparing this combination with sunitinib.
10. **b. Initiate therapy with a combination of nivolumab and ipilimumab since it is associated with better response rates and overall survival compared with sunitinib.** Nivolumab, in combination with ipilimumab, was superior to sunitinib in patients with previously untreated metastatic clear cell RCC. Sunitinib is no longer considered the standard of care in these patients. Lenvatinib plus everolimus has not been evaluated in this setting, and the combination of pazopanib and a PD1 inhibitor is deemed to have an unacceptable side effect profile.
11. **a. Axitinib.** In a randomized phase 3 trial, axitinib has been shown to prolong progression free survival compared with sorafenib in patients with metastatic clear cell RCC who have progressed on front-line therapy with either a VEGFR inhibitor or cytokine therapy.
12. **d. All of the above.** In separate randomized phase 3 studies, axitinib, in combination with either pembrolizumab or avelumab, was shown to be superior to sunitinib in previously untreated patients with metastatic clear cell RCC. Cabozantinib was associated with a higher response rate and PFS compared with sunitinib in patients with intermediate or poor-risk prognostic features in a randomized phase 2 study.
13. **d. Temsirolimus.** In a randomized phase 3 study, temsirolimus was associated with better overall survival than IFN-α (median 10.9 vs. 7.3 months) in metastatic RCC patients presenting with three or more predefined factors predictive of poor prognosis.
14. **e. Raf-1.** Raf-1 is a mediator of growth factor signaling pathways but has not been shown to be upregulated in RCC as a consequence of VHL inactivation/HIF upregulation.
15. **a. 70% to 90%. Based on numerous recent studies, it is estimated that VHL inactivation by mutation or promoter**

hypermethylation occurs in 70% to 90% of clear cell renal tumors.

16. **d. Enrollment in a phase 2 trial evaluating a novel inhibitor of MET activity is a reasonable consideration in this patient. There is no conclusive evidence suggesting that standard agents with activity in clear cell RCC (including VEGF pathway inhibitors and mTOR inhibitors) have a favorable impact on outcome in patients with metastatic papillary RCC, and patients with these tumors are appropriate candidates for rational targeted therapy approaches.** The presence of activating c-met mutations in some papillary tumors has kindled interest in the evaluation of *Met* pathway antagonists in this patient population.

Chapter Review

1. Two randomized studies have demonstrated improved survival in carefully selected metastatic RCC patients undergoing cytoreductive nephrectomy followed by cytokine therapy (IFN-α) compared with those receiving cytokine therapy alone.
2. Several patient and/or disease characteristics appear to influence outcome; patients with poor performance status, comorbid medical conditions, rapidly progressive disease, presence of brain metastases, and so on, are unlikely to benefit from this approach.
3. Although several retrospective studies suggested that cytoreductive nephrectomy may be of benefit in the era of VEGFR-targeted therapy, a randomized study failed to demonstrate an advantage in intermediate and poor prognosis patients. These data, however, do not specifically address the utility of nephrectomy in patients with relatively low metastatic burden and other favorable features, who may benefit from this approach.
4. It is our current practice to offer cytoreductive nephrectomy to well-selected patients with metastatic clear cell RCC
5. Resection of isolated metastatic lesions is appropriate in selected patients.

6. Several retrospective studies have suggested that patients undergoing complete resection of isolated metastatic foci may experience long disease-free intervals, with median overall survival rates of 35% to 50% in some reports.
7. Several factors are associated with an improved outcome after metastasectomy, including complete resection, presence of solitary metastatic lesions, age younger than 60 years, smaller tumor size, presence of pulmonary metastases, and development of metachronous metastatic disease.
8. There are no prospective, randomized studies demonstrating a favorable outcome with metastasectomy. It is therefore possible that the favorable outcome after resection of limited metastatic disease may be a reflection of patient selection bias, differences in tumor biology and natural history, or other confounding factors not related to resection.
9. In some patients with advanced RCC, cytoreductive nephrectomy may help alleviate symptoms related to the primary tumor (e.g., intractable pain, hematuria) or paraneoplastic manifestations.
10. However, nonsurgical options are often effective in palliating symptoms associated with RCC; cytoreductive nephrectomy is hence infrequently performed with purely palliative intent.
11. Resection of metastatic lesions (often in combination with radiation or systemic therapy) is sometimes performed for relief of symptoms or to prevent life-threatening or disabling sequelae.